

SIH 2026 — Team Briefing

Real-Time Airfare Price Index (APIx)

Problem Statement SIH26056 — Ministry of Statistics & Programme Implementation (MoSPI)

1. Problem Statement at a Glance

Field	Details
Title	Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the CPI
Organization	Ministry of Statistics and Programme Implementation (MoSPI)
PS Number	SIH26056
Category	Software
Theme	Travel & Tourism
Idea Submission Deadline	20 September 2026

2. Background — Why This Problem Exists

The Consumer Price Index (CPI), released by the National Statistical Office (NSO) under MoSPI, is India's primary measure of retail inflation. RBI uses CPI to set monetary policy under its flexible inflation-targeting framework.

Air travel fares fall under the CPI's 'Transport and Communication' sub-group, currently priced manually from a limited set of outlets — but over 90% of domestic tickets are bought online, and fares swing 200-400% in a single day based on advance-booking window, demand, season, and fuel surcharges. This project automates that data collection.

3. Who Our Client / Users Are

Primary users (explicitly named in the problem statement):

- MoSPI / NSO — would consume the index to augment CPI's transport sub-group
- RBI — downstream beneficiary via more accurate CPI feeding into monetary policy

Secondary / stretch beneficiary (our own addition, not a stated requirement):

- General travelers — e.g. 'cheapest booking window per route' as a bonus insight, always framed as an add-on, not a core requirement

4. What We ARE Building — Actual Current State

All four core components already exist in the workspace, and the build has gone further than the original Phase-1 plan intended at this stage:

Component	Current status
(a) Web-scraping engine	Built and ahead of plan: Scrapy + Playwright framework with 11 sources scaffolded; 2 spiders fully hardened — IndiGo (airline) and MakeMyTrip (OTA)
(b) Airfare database	Built — running on SQLite by default (Postgres supported via env-var swap, not currently active)
(c) Index-construction module	Built — cleaning, outlier handling, and weighted index computation all working, using hand-written Python (statistics module) rather than pandas/numpy
(d) Web dashboard + API	Both built — FastAPI backend, and a React + Vite + Recharts dashboard (4 pages) — dashboard was not meant to start until Phase 3

5. What We Are NOT Building — Verified Against Code

After the code review, this is the accurate list of what is genuinely still absent (not just deferred by plan):

- Booking/payment/PNR flow — never built, and not intended (matches original plan)
- Live CAPTCHA solving or bypass — the scraper only detects CAPTCHA walls and backs off (CaptchaBlockDetectionMiddleware); it does not defeat them (matches original plan)
- Active IP rotation in production — the code for it exists (ProxyRotationMiddleware) but ships disabled by default with no proxies configured; it only activates if someone sets HTTP_PROXY_LIST
- A forecasting model — no ARIMA, Prophet, or any forecasting model exists anywhere in the codebase. This is a genuine gap against the stated plan, not a deliberate scope cut
- Any deployment — nothing is deployed to Render, Railway, Vercel, or any host; no Dockerfile exists; everything runs only via local uvicorn/vite dev servers

6. Original MVP Plan vs. What Was Actually Built

The workspace did not follow the original phased, scoped-down MVP — it built much closer to the full end-state spec in one pass:

Item	Actual workspace state
Airline/OTA sources	11 sources scaffolded; 2 hardened spiders (1 airline + 1 OTA)
Routes	10 routes (full basket)
Advance-purchase windows	All 5 (T+1/7/15/30/45)
Scrape schedule	Once per day (cron-style)
Dataset source	Calibrated synthetic data generator
Dashboard	Already built (React/Vite/Recharts, 4 pages)
IP rotation	Infrastructure built, disabled by default

7. Build Phases — Status Update

Phase	Actual state
Phase 1 — data pipeline	Cleaning, index, and backtest all built and working. Used a synthetic data generator instead of a manually supplied file. No forecasting model exists — real gap.
Phase 2 — API	Built, matches plan
Phase 3 — dashboard	Already built now — out of sequence, but functionally complete
Phase 4 — live scraper	Framework already built ahead of schedule — 11 spiders plus scraping middlewares (including the inert proxy-rotation and CAPTCHA-detection logic)

8. Technical Architecture / Tech Stack — Plan vs. Actual

Layer	Actual
Scraping	Scrapy + Playwright (superset)
Cleaning / index math	Hand-written Python (statistics module) — same output, different implementation
Forecasting	Not implemented — gap
Database	SQLite by default (Postgres supported via DATABASE_URL, not active)
Backend / API	FastAPI — matches
Frontend	React + Vite + Recharts — matches (Vite, not Next.js)
Scheduling	APScheduler (in-process Python scheduler) — conceptually similar, not a literal system cron or cloud function
Deployment	Nothing deployed anywhere — no Dockerfile, no host configs, local dev servers only

9. Government Data Source for Training & Validation

Most parameters match the intended specification exactly:

Parameter	Plan vs. actual
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Base year	2024 — matches
Series	Current — matches
Sector	Combined — matches
Division / Group / Class / Sub-class / Item	Transport and Communication → Passenger transport services → Passenger transport by air → domestic → Airfare — matches exactly
Imputation	N — matches
State	Plan called for "All India"; the actual data file (cpi_1054.xlsx) contains only 5 individual states — Karnataka, Maharashtra, Tamil Nadu, Telangana, West Bengal. No All-India row exists in the file at all.

10. Validation — Plan vs. Actual

The problem statement requires at least 30 days of back-tested results. The workspace currently delivers 181 days (about 6 months) of backtest coverage — exceeding the minimum ask. Two caveats apply, both already documented in docs/VALIDATION.md:

- The backtest runs on synthetic data, not live-scraped fares (consistent with Phase 1 not yet having live scraping wired in)
- The comparison baseline is the 5-state proxy described above, not the true All-India CPI airfare series

11. Known Gaps — Things That Genuinely Need Fixing

These are not scope decisions — they are gaps against what was actually planned:

- No forecasting model (ARIMA/Prophet or otherwise) exists anywhere in the codebase, despite being part of the Phase 1 spec
- CPI validation compares against a 5-state proxy average, not the true All-India series, because the supplied data file has no All-India row — needs either a genuine All-India data source or explicit, honest disclosure in the pitch
- Nothing is deployed — if the demo needs a live/hosted link rather than local dev servers, deployment work is still outstanding